UKUDLA Machine Learning Certificate

Introduction to Machine Learning for Agri-Food Systems

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UKUDLA
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for Sustainable and Resilient Food Systems and Applied Agricultural and Food Data Science

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Course Objectives

By the end, you will be able to

- Recognise when ML adds value to an agri-food question
- Distinguish predictive from causal modelling
- Select appropriate algorithms for common data types
- Implement a basic ML pipeline in Python
- Critically evaluate model results and avoid pitfalls

Domain expertise + ML skills = transformative innovation.

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Session Roadmap

Week	Focus	Key topics
1	Foundations	What is ML, T/E/P, **predictive vs causal**
2	Data Preprocessing	Data analysis, feature engineering, EDA
3	Regression	Linear, polynomial, decision trees, evaluation
4	Classification	Logistic, KNN, SVM, tree ensembles, metrics
5	Unsupervised + Evaluation	Clustering, PCA, bias–variance, tuning
6	Neural Networks & Vision	MLP, CNNs, transfer learning for agri imagery
7	Integration & Capstone	Cross-method synthesis, project clinic

A Story from the Field

- A smallholder farmer loses 30% of their maize harvest to a disease they noticed *too late*.
- A coffee cooperative cannot predict which batches will fail quality checks — until after shipment.
- A dairy farm wastes 20% of feed because rationing is based on experience, not data.

What if...

> ...these decisions could be informed by *data* and *machine learning* — **before** the loss occurs?.

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Data Explosion

Agriculture generates **terabytes** of data daily.
Data are generated daily through different sources such as:

- farm management records
- IoT sensors
- genomic
- market data

How do we make sense of this huge volume of data?

> **Domain expertise + machine learning skills = transformative innovation.**

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The Gap We Are Here to Close

There is a critical gap between data collection and actionable decision-making in agriculture [1].

What we have:

Sensors, satellites, drones, farm records — a torrent of data.

What we lack:

The methods to turn that data into predictions, decisions, and impact.

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Machine Learning

Machine learning (ML) is concerned with software, algorithms and systems that learn from data.

What is Machine Learning?

A computer program is said to **learn** from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, **improves with experience** E [2].

Example: Training a program to detect whether a maize leaf in a field image is *diseased* or *healthy*.

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Same framework applies to yield prediction, pest detection, food quality grading, and more.

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Pick a problem from your own research.

What are

T,

E,

P?

Predictive vs Causal Modelling

Two fundamentally different questions.

Predictive question

"What will happen?"

Goal: *Minimise prediction error on new data.*

Data: *Any correlated features will do.*

Example: *Forecast maize yield from weather + soil.*

Causal question

"What happens if we intervene?"

Goal: *Estimate the effect of a treatment or action.*

Data: *Requires design (RCT) or assumptions to handle confounding.*

Example: *Effect of no-till on yield.*

ML excels at **prediction**. It does *not* estimate causal effects by default — no matter how accurate the model.

Why the Distinction Matters

Classify each question — predictive or causal?

Forecasting tomorrow's soil moisture.
Predictive

Yield gain from switching to no-till.

Detecting diseased leaves in drone imagery.

Effect of an extension programme on technology adoption.

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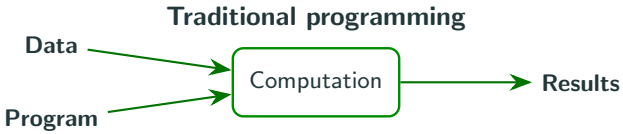
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Causal

Why it matters: Most agricultural data is observational (surveys, satellite time series) and confounded — wealth, location, and management all co-vary.

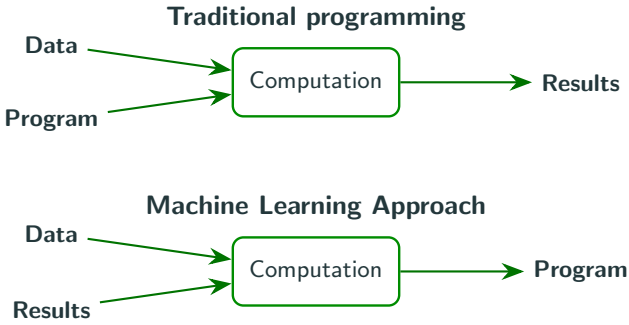
This course teaches predictive modelling. Causal inference requires additional design-based methods, it is not the focus here.

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Traditional Programming vs ML

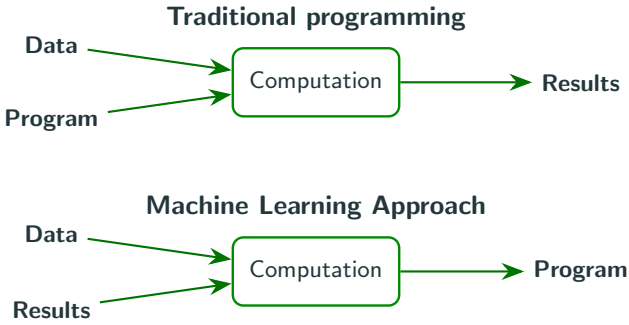


Traditional Programming vs ML



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Traditional Programming vs ML



The shift: We no longer write the rules. We supply data and outcomes, and the machine *derives* the rules.

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Why now? Not 30 years ago? What changed?

1. Faster hardware

GPUs and TPUs make large-scale computation affordable.

2. Improved algorithms

Better architectures, optimisers, and regularisation.

3. Explosion of data

Sensors, satellites, smartphones — agriculture included.

4. Excellent frameworks

scikit-learn, TensorFlow, PyTorch — accessible to non-specialists.

As earlier noted: Agriculture is now one of the most data-rich sectors on Earth.

Where is ML Already Working?

Let us brainstorm (60 seconds): Call out any application of ML you have heard of.

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- Financial trading
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(Google reduced data-centre electricity use by **15%** *using ML [3].)*

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And in Agri-Food Systems?

Where could ML help *your* research?

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Production

- Crop yield prediction
- Pest & disease detection
- Irrigation scheduling
- Soil fertility mapping

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- Supply chain optimisation
- Post-harvest loss prediction
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This course is where your domain expertise meets ML methods.

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Three Families of Machine Learning

Supervised

Inputs *with* desired outputs.
The model learns to mimic patterns.

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Quick Check

Which category fits these agri-food problems?

- Predicting maize yield from weather and soil
- Grouping farms by soil similarity
- A robot learning to pick ripe fruit

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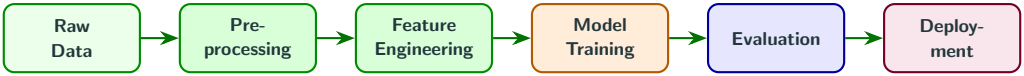
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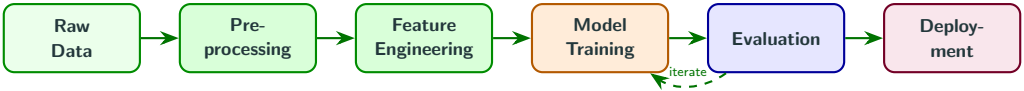
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The Machine Learning Workflow



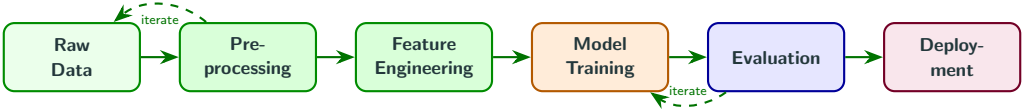
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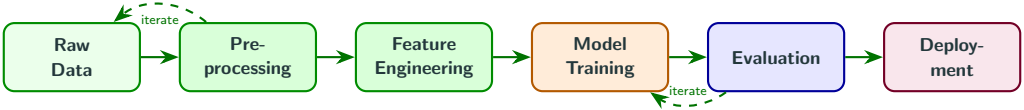
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The Machine Learning Workflow



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The Machine Learning Workflow



Key insight: ML is not a linear process. Preprocessing and evaluation feed back into earlier stages — often many times.

Activity: Map It!

Choose one agri-food problem from the list below.
Sketch the ML workflow for it:

raw data → preprocessing → features → model → evaluation → deployment.

Choose one

- Crop yield prediction
- Disease detection from leaf images
- Food demand forecasting
- Soil type clustering

Guiding questions

- What data do you have?
- What does T, E, P look like?
- Supervised, unsupervised, or RL?
- What could go wrong?

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Key Takeaways

1. ML is about *learning from data*, not about following hard-coded rules.

2. Mitchell's T, E, P framework applies to every ML problem — including yours.

3. Three families: supervised, unsupervised, reinforcement.

4. ML is a *workflow*, not a single tool — data, features, model, evaluation, deployment.

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